High-speed
Centrifuging Spray dryer

Use Manual

Working principle and Features

After the air passes through the filter and the heating device, the hot air distributor enters the top of the drying chamber, and the hot air snail through the hot air distributor rotates into the drying chamber uniformly, and the pump sends the liquid to the top of the drying chamber. The spray head, the liquid is sprayed into extremely small droplets, and the specific surface area of the liquid contact with the hot air is greatly increased. The small droplets and the hot air collide and sink together. The water evaporates quickly and is dried into pellets in a very short period of time. It is trapped in the bottom of the tower and the cyclone, and the exhaust gas is discharged to the outside.

1. Install the spray dryer to correct the level, put the drying tower in the appropriate position, install the escalator and railing, work platform, fan and motor, cyclone and air heater according to the equipment installation diagram .
2. Turn on the power and install the electrical cabinet in a convenient location. In order to ensure safe production, the ground wire must be installed to install the indus and electrical control ground.
3. The exhaust outlet of the centrifugal fan should be equipped with a 1:4 exhaust duct according to the installation site, so that the exhaust gas can be taken out of the room without increasing the indoor humidity and reducing the noise in the workshop.

Using

In order to ensure the quality of the product, it must be fully cleaned before use, and should be disinfected according to the requirements of the product.

1. **Check the following before use:**

①The sealing material is installed at the pipe joint, and then the gas is flushed to ensure that unheated air is not allowed to enter the drying chamber.

Close the

② door and observation window and check for air leaks.

The bottom of the barrel and the pollinator at the bottom of the cyclone should be checked for leaks before installation.

③ The pollinator can be tightened before it is removed, and the pollinator must be cleaned and dried.

④ Check that the rotation direction of the centrifuge is correct.

⑤ The adjustment butterfly valve at the outlet of the centrifugal fan is opened, and the butterfly valve is closed, otherwise the electric heater and the inlet pipe will be damaged. This must be paid enough attention.

⑥ Whether the connecting pipe of the feed pump is connected, the direction of rotation of the motor and the pump is correct.

⑦ Place the spray head on the top of the drying chamber to prevent air leakage.

2、working

① First turn on the centrifugal fan, then turn on the electric heating, and check whether there is air leakage. If it is normal, the barrel body can be preheated. The hot air preheating determines the drying evaporation capacity, without affecting the quality of the dried material. The inlet air temperature should be increased as much as possible.

② When preheating, place the spray head on the top of the drying chamber. The bottom of the drying chamber and the outlet of the cyclone separator must be closed to prevent cold air from entering the drying chamber and reduce the preheating efficiency.

③ When the inlet humidity of the drying chamber reaches 150°C～200°C, open the centrifugal nozzle. When the spray head reaches the set speed, turn on the feed pump and add the feed liquid. The amount of the feed should be small to large, otherwise the sticky wall will be generated. Phenomenon until adjusted to the appropriate requirements. The concentration of the liquid should be prepared according to the nature of the dryness of the material to ensure good fluidity of the finished product after drying.

④ The temperature and humidity of the dried product depend on the exhaust air temperature. It is extremely important to keep the exhaust air temperature constant during operation. It depends on the amount of material to be discharged. The material volume is stable and the outlet temperature is not changed. of. The outlet temperature also changes if the solids content and flow rate of the feed liquid changes.

⑤ The product temperature is too high, which can reduce the amount of feed to increase the outlet temperature. The humidity of the product is too low, otherwise it is the opposite. For heat sensitive materials with lower product temperatures, the amount of feed can be increased to reduce the exhaust air temperature, but the temperature of the product will increase accordingly.

⑥ Collection of finished products after drying. In the lower part of the tower body or in the lower part of the cyclone separator, the receiving barrel should be exchanged before it is filled. When the pollinator is replaced, the upper butterfly valve must be turned off before proceeding.

⑦ If the dried product is hygroscopic, the cyclone separator and its pipe and the pollination tube are covered with heat-insulating material, so as to avoid moisture respiration of the dried product.

3,Cleaning

1 In order to ensure the cleaning of the drying tower body and its pipes and all the parts in contact with the finished product, it is necessary to obtain first-class products and regular cleaning equipment.

2 When the product is replaced, or if the equipment is shut down for more than 24 hours without being cleaned, a thorough and thorough cleaning should be carried out.

3 washing methods:

Dry cleaning: Clean with a brush, broom and vacuum cleaner.

Wet wash: Wash with hot water at 60-80 °C.

Chemical washing: Wash with alkali, acid and various detergents.

Pickling: Nitric acid (NHO3) is formulated into a 1 to 2% solution, and the heating is carried out at a temperature not exceeding 65 ° C, followed by washing with water.

Alkaline washing: sodium hydroxide (NaOH) is formulated into a 0.5 to 1% solution, heated to 60-80 ° C for washing, and then washed with water.

4 After the wet and chemical washing, the equipment and various components should be installed and subjected to high temperature sterilization for 15 to 30 minutes.

5 When cleaning the equipment, care should be taken not to wash with chlorine or other compounds.

6 Air filter cleaning should be based on the surrounding environmental conditions, that is, the amount of dust in the air. It is cleaned once every 3 to 6 weeks, and cleaned once every 6 to 8 weeks with low dust content.

After the stainless steel filaments in the seventh stage are washed by the alkali washing method, they should be evenly laid in the frame of the air filter, and sprayed with the spindle oil and the vacuum pump oil.

Maintenance

Maintenance of the centrifugal nozzle is particularly important in order to operate smoothly at high speeds.

1. If there is noise and vibration during the use of the nozzle, stop the nozzle immediately and check if there is residual material in the spray tray. If any, it should be cleaned in time.
2. Check whether the bearing and bushing and the transmission parts such as shaft and gear are abnormal. If it is found, the ring parts should be replaced in time.
3. The mechanical transmission nozzle is rotated by high-speed gear. It must be cooled and cooled by high-speed lubrication and cooling oil, so that the gears, shafts and bearings are well lubricated (hydraulic oil or spindle oil can be used). The viscosity of the use should not be too high.
4. In order to increase the service life of the nozzle, it is best to use the nozzle alternately for 8 hours or according to the situation. The lubricating oil of the rolling bearing should be exchanged once every 150

to 200 hours. In the use of the oil cup switch every few 1-2 hours, add a few drops of lubricating oil lubrication bushing. The oil in the oil cup on the top of the oil cup will be quickly added, and the product will be contaminated due to too much fuel.

5. After use, the spray disc should be removed, immersed in water, and the residual material washed with water. Brush it with a brush when it is not washed with water. Because the residual material on the spray disk will bring the imbalance of the spray disk, which will seriously affect the service life of the spray head and even damage other parts of the ring.

6. When disassembling the spray head, care should be taken to bend the main shaft. When installing the spray disc, use the gap between the plug control disc and the housing. The nut of the fixed spray disc must be tightened to prevent loosening.

7. Do not lay down after the nozzle is finished and during transportation. Improper placement will bend the spindle and affect the use. Therefore, there should be a fixed frame.

Analysis of problems that may occur in production

In the production process, sometimes abnormal phenomena occur, especially new equipment, the first operation, when the operation is not skilled, the lack of experience in adjustment control, it is more likely to cause some problems, and now there may be problems and causes

Please see the following table

Problem	Reason	Solution
High water content	The exhaust air temperature is too low.	Properly reduce the amount of feed to increase the exhaust air temperature.
There is sticking phenomenon everywhere in the interior wall of the drying room	1. The feed amount is too large to evaporate sufficiently.	1. Appropriately reduce the amount of feed.
	2. The drying chamber is not heated enough before the spray starts.	2. Properly increase the temperature of import and export.
	3. The flow rate of the feed is too large when the spray is started.	3. When starting the spray, the flow rate should be small, gradually increase, and adjust to appropriate.
	4. The added feed liquid is not stabilized or the feed amount is too large or too small.	4. A. Check if the pipeline is blocked.
		B. Adjust the solid content of the material to ensure the fluidity of the liquid.

Reduced evaporation	1. The amount of air in the entire system is reduced. 2. The inlet air temperature of hot air is low. 3. The equipment has a phenomenon of slippage, and wind enters the drying room.	1. Check if the speed of the centrifugal fan is normal. 2. Check if the position of the butterfly valve at the outlet of the centrifugal fan is correct. 3. Check if the air filter and heater pipe are blocked. 4. Check if the voltage is working properly. 5. Check if the electric heater is working properly. 6. Check whether the joints of the components of the equipment are sealed. 7, check whether the steam pressure is reduced。
Excessive impurities in finished products	1. The air filter is not effective. 2. The powder is mixed into the finished product. 3. The purity of the feed liquid is not high. 4, equipment cleaning is not complete。	1. Check if the stainless steel mesh in the air filter is evenly distributed and the oil is dry. 2. The filter is used for too long and should be replaced immediately. 3. Check the coke powder at the entrance to the hot air and turbulence. 4. Filter the liquid before spraying. 5. Re-clean the equipment.
Product powder is too fine	1. The solid content of the feed liquid is too low. 2. The amount of feed is too small.	1. Increase the solid content of the feed liquid. 2. Increase the feed amount and increase the inlet air temperature accordingly.
Centrifugal nozzle speed is too low	Centrifugal nozzle parts are faulty	Stop using the nozzle to inspect the parts inside the nozzle.
There is vibration when the centrifugal nozzle is running	1. There is residual material on the spray plate. 2, shaft generation	1. Inspect and clean the spray tray. 2, replace the new shaft。